

SUMMARY

Experienced, pragmatic, and solutions-driven cybersecurity engineer with extensive experience in cybersecurity, information security, and IT operations. Known to build strong customer relationships and deliver value-added solutions.

AREAS OF EXPERTISE

Infrastructure Security - DevOps & Automation - System Administration - Threat Modeling - Risk Analysis

TECHNICAL SKILLS

Project Management: Trello, Jira

Languages: Python, Java, C++, SQL

SIEM: Elastic Stack, Splunk

Virtualization: KVM, VMware vSphere, Proxmox

Infrastructure as Code: Terraform, Ansible, CloudFormation

System Administration: Red Hat Enterprise and Ubuntu Server

Version Control Systems: GitLab, GitHub

Developer Tools: VS Code, Eclipse, PyCharm

Containerization: Docker, Kubernetes, Portainer

Monitoring Solutions: vROps, vRNI, Zabbix

Cloud: AWS, OpenNebula

Configuration Management: NetBox

WORK EXPERIENCE

Career Break

June 2023 – Present

- Took a planned career break to prioritize health and recovery.
- Continued professional development through certifications and self-study.
- Now actively seeking to re-enter the workforce with fresh perspective and skills.

Mastercard

February 2022 – June 2023

Lead Information Security Engineer, Cyber Range

Arlington, VA

Mastercard cyber-range is a simulated environment built on top of AWS cloud and VMware vSphere. In this role, I was responsible for building, deploying, automating, configuring, and securing large-scale physical and virtual infrastructures to support our customers. The role is a mixture of DevSecOps, System Administration, and Network Security.

- Executed CTF and CDX competitions with 80% customer satisfaction rate.
- Implemented Elastic Stack for centralized logging, enhancing threat detection and response..
- Deployed Zabbix for system performance monitoring, ensuring 99.99% up-time.
- Reduced support tickets by 47% through an automated alerting mechanism for system health.
- Automated infrastructure deployment with Terraform, reducing manual effort by 87%.
- Build and maintain AWS VPC, VPC Gateways, S3 buckets, EC2 instances, reusable modules, and IAM accounts using Terraform.
- Wrote ansible playbooks to install applications, apply patches, and update software on Windows and Linux systems.
- Implemented capacity planning in the data center to accommodate 50% influx in resource demand.
- Proactively spearheaded VM efficiency project in the cyber-range, leading to \$700 monthly OPEX reduction.
- Provided technical support to different business units, fostering collaboration while resolving diverse technical issues.
- Maintained GitLab repositories and conducted code reviews for security and performance improvements.
- Created and maintained information security standard, operating procedure, and architecture documents related to cyber-range.

Mastercard

September 2019 – January 2022

Lead Information Security Engineer, Security Research

Arlington, VA

In this role, I was responsible for leading complex initiatives to advance our security program and conduct cutting-edge research to keep Mastercard safe from current and future threats.

- Conducted cutting-edge security research, and provide strategic guidance fit to upper management.
- Researched the latest security trends, identified security gaps and recommended security solutions to different business units.
- Conducted Proof of Concepts (PoCs) to evaluate and implement new security technologies.
- Led a team of six to develop a real-time attack visualization solution for cyber defense competition.
- Spearheaded cybersecurity education initiatives, enhancing security awareness across Mastercard.

Dakota State University

August 2016 – August 2019

Instructor of Cybersecurity

Madison, SD

- Taught web security and network security to graduate and undergraduate students.
- Provided hands-on training using black box approach to identify and exploit bugs on web applications.

Garrett College

August 2015 – May 2016

Instructor of Cybersecurity

McHenry, MD

- Established a cybersecurity lab and trained faculty on security best practices..
- Educated management on cyberattack prevention and mitigation strategies.

Towson University

August 2009 – December 2014

Research Assistant

Towson, MD

- Developed an attack framework for simulating cyberattacks on power grids and telecom networks..
- Conducted penetration testing to identify vulnerabilities in cyber-physical systems.

SELECTED PUBLICATIONS

IoT security in payments space | *Mastercard Internal Publication*

- Identified the risk of integrating payments with IoT devices.
- Proposed remediation strategies to accelerate the adoption of IoT in payments space.

Benefits of 5G to payments ecosystem | *Mastercard Internal Publication*

- Proposed the benefits of 5G to payments space.
- Evaluated 5G security risks and standardization challenges.

On simulation studies of cyber attacks against LTE network | *IEEE ICCCN*

- Proposed a comprehensive threat model of LTE network.
- Simulated cyber-attacks on LTE network based on realistic network topology.

Data manipulation attack in smart grid | *Security & Communication Network*

- Developed an integrated technique to detect meter data manipulation attacks in smart grid.
- The proposed technique was found to combat both subtle and strong manipulation of data at low cost.

EDUCATION

Towson University

PhD in Information Technology

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Certified Kubernetes Administrator(CKA)
- CompTIA Security+
- CompTIA Network+
- Graduate Certificate in Leadership Development